Module - 2

05/02/26

Analysis of Searching and Sorting Algorithms

I
1. Linear search
2. Binary Search

II
1. Quick sort
2. Merge sort
3. Heap sort
- Comparison of sorting algorithms

III Analysis of complex Data structures
1. Binomial heap
2. Fibonacci heap
3. Red - black tree

Binary search

* Following divide and conquer method

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
3	6	8	12	14	17	25	29	31	36	42	47	53	55	62
l							mid	l	mid	l	n			h

Case 1:

Key = 42

l	h	mid = $\frac{l+h}{2}$
1	15	8
9	15	12
9	11	10
11	11	11

→ 4 searches

⇒ mid = key
42 = 42

Case 2: Key = 17

l	h	mid
1	15	8
1	7	4
5	7	6

⇒ mid = key
17 = 17

3 searches

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case 3 : key = 55

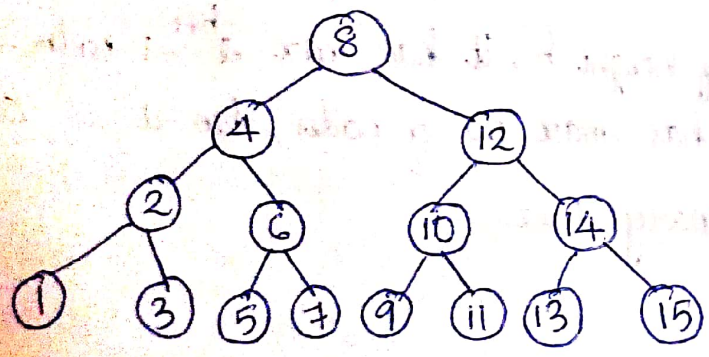
l	h	$mid = \frac{l+h}{2}$
1	15	$\frac{1+15}{2} = 8$
9	15	$\frac{9+15}{2} = 12$
13	15	$\frac{13+15}{2} = 14$

$\Rightarrow$ mid = key
55 = 55

3 searches.

```
int BinSearch (A, n, key)
{
    l = 1;
    h = n;
    while (l ≤ h) {
    mid = (l+h)/2;
    if (A[mid] = = key)
        return mid
    if ( key > A[mid])
        l = mid + 1;
    else
        h = mid - 1;
    }
    return 0;
}
```

Tree.



Max no. of searches req = Height of the tree
= $\log n$

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worst case = $O(\log n)$

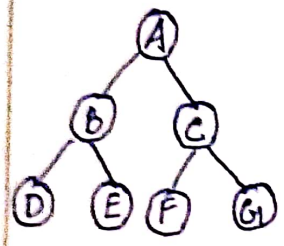
Best case = $O(1)$

Average = $O(\log n)$

10/02/20 **Assignment**

Quick sort
Merge sort } eg. Algm, Analysis (Best, Avg, worst)

Binary Tree and its Representation

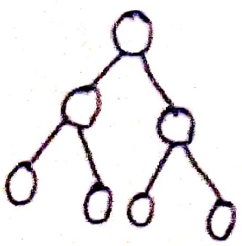


If a node is at index i
left child - $2*i$
right child - $2*i+1$
parent node - $i/2$

A	B	C	D	E	F	G
1	2	3	4	5	6	7

Eg: B ⇒ left child $= 2*2 = 4$ — D
right child $= 2*2+1 = 5$ — E
parent node $= \frac{i}{2} = \frac{2}{2} = 1$ — A

Full Binary tree



A tree of height h, it has max $2^{h+1}-1$ nodes.
If it has max no. of nodes, then it is called full binary tree.

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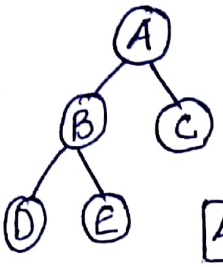
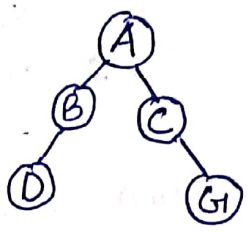
Complete Binary tree



⇒ leaf nodes are at level h or h-1. leaf nodes are filled from left to right and the array will not have any gaps or free slots.

A	B	C	D	E

A complete binary tree of height h is a full binary tree upto h-1 and in the last level the elements are filled from left to right



A	B	C	D	–	–	G

↳ not complete binary tree

Heap → It is a complete binary tree

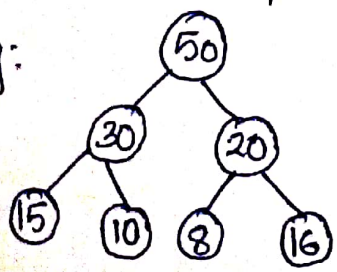
- Max heap
- Min heap

* For a max heap, every parent node is having a value >= all its descendents or child nodes.

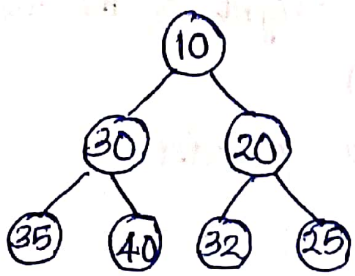
* For a min heap, ~~eg~~ every parent node is having a value <= all its child nodes.

eg:

Max heap



Min heap



Heap sort - heapify

combination of insertion & deletion.

1. insertion 2. deletion

heapify means to satisfy the properties of min heap or max heap.

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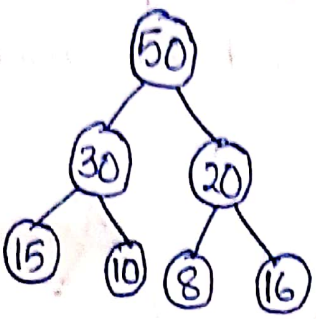
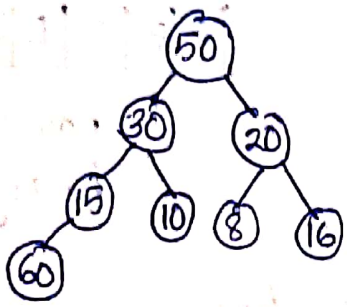
Insertion of an element into a heap

Max heap



50	30	20	15	10	8	16
1	2	3	4	5	6	7

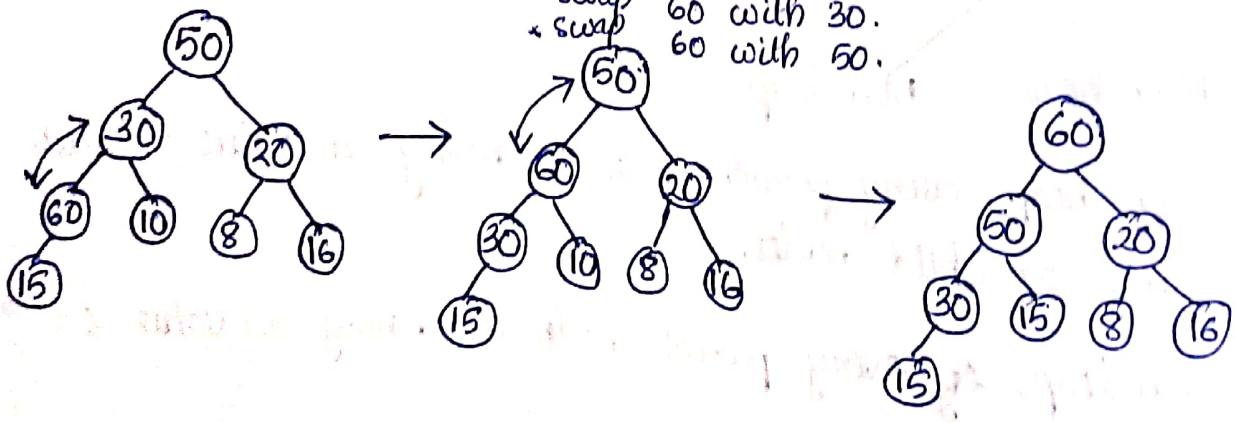
Insertion of 60



50	30	20	15	10	8	16	60

→ insert 60 as a child node

→ heapify
- swap 60 with 15.
- swap 60 with 30.
- swap 60 with 50.



no. of swaps required = height of the tree.

$T(n) = O(\log n)$

where $n \rightarrow$ no. of nodes